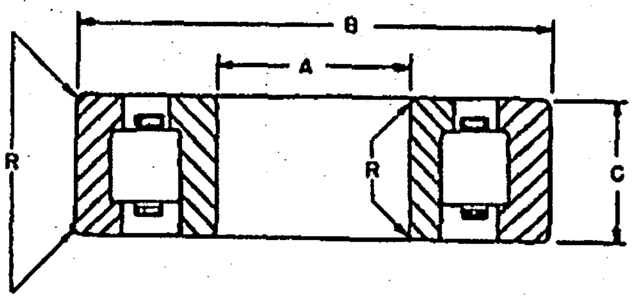
This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

NOTICE—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

FED. SUP CLASS
3110



THE CORNER RADIUS OR CHAMFER ON BOTH SIDES OF THE BEARING MUST CLEAR THE SHAFT AND HOUSING FILLET RADIUS LISTED IN THE TABLE BELOW

MS PART NO	FIIN	A INSIDE DIA		B OUTSIDE DIA		C WIDTH		R SHAFT & HOUSING FILLET RADIUS	RECOMMENDED SHOULDER DIAMETER SHAFT	RECOMMENDED SHOULDER DIAMETER HOUSING	INTERNAL RADIAL CLEARANCE .0001 INCH		BASIC DYNAMIC LOAD RATING	BASIC STATIC LOAD RATING	RECOMMENDED LIMITING SPEED	EST WEIGHT
		MM	INCH	MM	INCH	MM	INCH	MAX	MIN	MAX	MIN	MAX	LBS	LBS	RPM	LBS
MS19078 - 1		17	.6693	40	1.5748	12	.4724	.025	.84	1.44	4	16	2500	1400	8750	.19
MS19078 - 2		17	.6693	47	1.8504	14	.5512	.040	.88	1.67	4	16	2900	1500	7500	.32
MS19078 - 3		20	.7874	52	2.0472	15	.5906	.040	1.02	1.82	4	16	3500	1900	6250	.37
MS19078 - 4		25	.9843	52	2.0472	15	.5906	.040	1.22	1.88	4	18	3000	1600	6500	.35
MS19078 - 5		25	.9843	62	2.4409	17	.6693	.040	1.25	2.20	4	18	5100	2900	6000	.64
MS19078 - 6		30	1.1811	62	2.4409	16	.6299	.040	1.42	2.24	4	18	3850	2200	5500	.57
MS19078 - 7		30	1.1811	72	2.8346	19	.7480	.040	1.50	2.55	4	18	6750	3050	5000	.83
MS19078 - 8		35	1.3780	72	2.8346	17	.6693	.040	1.68	2.61	6	20	6000	3450	4700	.78
MS19078 - 9		35	1.3780	80	3.1496	21	.8268	.060	1.73	2.83	6	20	8350	5350	4350	1.20
MS19078 - 10		40	1.5748	80	3.1496	18	.7087	.040	1.87	2.88	6	20	7650	4850	4200	.93
MS19078 - 11		40	1.5748	90	3.5433	23	.9055	.060	1.94	3.19	6	20	10300	6350	3650	1.74
MS19078 - 12		45	1.7717	85	3.3465	19	.7480	.040	2.03	3.06	8	22	9050	5600	3350	1.15
MS19078 - 13		45	1.7717	100	3.9370	25	.9843	.080	2.20	3.54	8	22	14200	9050	3450	2.32
MS19078 - 14		50	1.9685	110	4.3307	27	1.0630	.080	2.42	3.88	8	22	17600	11150	3150	3.04
MS19078 - 15		55	2.1654	100	3.9370	21	.8268	.060	2.52	3.60	8	26	10300	6700	3200	1.71
MS19078 - 16		55	2.1654	120	4.7244	29	1.1417	.080	2.65	4.25	8	26	20700	13500	2850	3.65
MS19078 - 17		60	2.3622	110	4.3307	22	.8661	.080	2.73	3.96	8	26	12600	8450	2900	2.23
MS19078 - 18		60	2.3622	130	5.1181	31	1.2205	.080	2.83	4.61	8	26	23200	15500	2600	4.32
MS19078 - 19		65	2.5591	120	4.7244	23	.9055	.060	3.00	4.32	8	26	14900	10200	2700	2.83
MS19078 - 20		65	2.5591	140	5.5118	33	1.2992	.080	3.11	4.96	8	28	25600	17100	2400	5.83
MS19078 - 21		70	2.7559	125	4.9213	24	.9449	.080	3.19	4.50	10	30	14800	10200	2550	3.32
MS19078 - 22		70	2.7559	150	5.9055	35	1.3780	.080	3.32	5.31	10	30	30100	20300	2250	7.47
MS19078 - 23		75	2.9528	130	5.1181	25	.9843	.050	3.37	4.68	10	30	17000	12100	2400	3.22
MS19078 - 24		75	2.9528	160	6.2992	37	1.4567	.080	3.57	5.67	10	30	33300	25600	2100	8.55
MS19078 - 25		80	3.1496	140	5.5118	26	1.0236	.060	3.63	5.05	10	30	19600	13700	2250	3.82
MS19078 - 26		80	3.1496	170	6.6929	39	1.5354	.080	3.70	6.02	10	30	36500	26600	1950	10.47
MS19078 - 27		85	3.3465	150	5.9055	28	1.1024	.080	3.85	5.40	12	32	22400	15900	2100	5.00
MS19078 - 28		85	3.3465	180	7.0866	41	1.6142	.100	4.04	6.38	12	32	42900	30500	1850	12.26
MS19078 - 29		90	3.5433	160	6.2992	30	1.1811	.080	4.05	5.78	12	32	28000	20200	1950	5.98
MS19078 - 30		90	3.5433	190	7.4803	43	1.6929	.100	4.25	6.73	12	32	46500	33300	1750	14.21
MS19078 - 31		95	3.7402	170	6.6929	32	1.2598	.080	4.29	6.12	12	32	31400	22000	1850	7.18
MS19078 - 32		95	3.7402	200	7.8740	45	1.7717	.100	4.54	7.09	12	32	51200	33100	1650	16.94
MS19078 - 33		100	3.9370	180	7.0866	34	1.3386	.080	4.56	6.40	12	32	34000	25000	1750	8.62

APPROVED 30 APR 1959 REVISED

P.A. None
Other Cust
ORD
USAF

TITLE
BEARINGS, ROLLER, CYLINDRICAL, SINGLE ROW OF ROLLERS, ONE ROLLER RETAINING RIB ON INNER RING, TWO ROLLER RETAINING RIBS ON OUTER RING, ONE DIRECTION LOCATING.

PROCUREMENT SPECIFICATION
FF-B-185

SUPERSEDES:

MILITARY STANDARD

MS 19078

SHEET 1 OF 3

DD FORM 672-1

This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

NOTICE.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

FED. SUP CLASS
3110

MS PART NO	FIIN	A INSIDE DIA		B OUTSIDE DIA		C WIDTH		R SHAFT & HOUSING FILLET RADIUS	RECOMMENDED SHOULDER DIAMETER SHAFT	RECOMMENDED SHOULDER DIAMETER HOUSING	INTERNAL RADIAL CLEARANCE .0001 INCH		BASIC DYNAMIC LOAD RATING	BASIC STATIC LOAD RATING	RECOMMENDED LIMITING SPEED	EST WEIGHT
		MM	INCH	MM	INCH	MM	INCH	MAX	MIN	MAX	MIN	MAX	LBS	LBS	RPM	LBS
MS19078 - 34		100	3.9370	215	8.4646	47	1.8504	.100	4.83	7.62	12	32	61500	45800	1550	19.50
MS19078 - 35		105	4.1339	190	7.4803	36	1.4173	.080	4.77	6.64	14	35	38000	30000	1500	10.32
MS19078 - 36		105	4.1339	225	8.8583	49	1.9291	.100	5.05	7.97	14	35	69700	52400	1450	22.95
MS19078 - 37		110	4.3307	200	7.8740	38	1.4961	.080	5.01	7.20	14	35	45000	34000	1550	11.01
MS19078 - 38		110	4.3307	240	9.4488	50	1.9685	.100	5.37	8.50	14	35	79100	60300	1350	27.38
MS19078 - 39		120	4.7244	215	8.4646	40	1.5748	.080	5.47	7.74	14	35	49500	37520	1450	14.70
MS19078 - 40		120	4.7244	260	10.2362	55	2.1654	.100	5.82	9.21	14	35	93400	71100	1150	35.00
MS19078 - 41		130	5.1181	280	11.0236	58	2.2835	.120	6.32	9.92	18	41	110000	90100	850	43.55
MS19078 - 42		150	5.9055	320	12.5984	65	2.5591	.120	7.11	11.34	18	45	133000	106000	950	62.00

APPROVED 30 APR 1958 REVISED

P.A. NORD
Other Cust
ORD
USAF

TITLE
BEARINGS, ROLLER, CYLINDRICAL, SINGLE ROW OF ROLLERS, ONE ROLLER RETAINING RIB ON INNER RING, TWO ROLLER RETAINING RIBS ON OUTER RING, ONE DIRECTION LOCATING.

MILITARY STANDARD

MS 19078

PROCUREMENT SPECIFICATION
FF-B-185

SUPERSEDES:

SHEET 2 OF 3

DD FORM 1 SEP 57 672-1

This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

NOTICE.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

FED. SUP CLASS
3110

TOLERANCES: STANDARD GRADE (TOLERANCES IN .0001 INCH).

INNER RING				
BORE MM		A		RADIAL RUNOUT
OVER	INCL	MIN	MAX	MAX
0	10	- 3	+0	3
10	18	- 3	+0	4
18	30	- 4	+0	5
30	50	- 5	+0	6
50	80	- 6	+0	8
80	120	- 8	+0	10
120	180	-10	+0	12
180	250	-12	+0	16

OUTER RING				
OD MM		B		RADIAL RUNOUT
OVER	INCL	MAX	MIN	MAX
0	18	+0	- 4	6
18	30	+0	- 4	8
30	50	+0	- 5	8
50	80	+0	- 5	10
80	120	+0	- 6	14
120	150	+0	- 8	16
150	180	+0	-10	18
180	250	+0	-12	20
250	315	+0	-14	24
315	400	+0	-16	28

WIDTH			
NOMINAL BORE MM		C	
OVER	INCL	MAX	MIN
0	180	0	50

THE TOLERANCES FOR WIDTH APPLIES TO INDIVIDUAL RINGS.

MATERIAL: RINGS AND ROLLERS; ALLOY STEEL, FED. STD. 66 STEEL ID'S., 4320, 4620, 8620 OR 52100. CAGE; BRASS, BRONZE OR STEEL (MACHINED).

HARDNESS: RINGS AND ROLLERS; 58 RC. TO 65 RC.

DIMENSIONS: UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.

LUBRICATION: THE IDENTIFYING NUMBER DOES NOT INCLUDE LUBRICATION.

BASIC DYNAMIC LOAD RATING: BASIC DYNAMIC LOAD RATING IS THAT CONSTANT STATIONARY RADIAL LOAD WHICH A GROUP OF APPARENTLY IDENTICAL BEARINGS WITH STATIONARY OUTER RING CAN ENDURE FOR A RATING LIFE OF ONE MILLION REVOLUTIONS OF THE INNER RING.

RATING LIFE (HOURS): RATING LIFE IS THE NUMBER OF HOURS AT SOME CONSTANT SPEED OF THE INNER RING THAT 90 PER CENT OF A GROUP OF APPARENTLY IDENTICAL BEARINGS WILL COMPLETE OR EXCEED BEFORE THE FIRST EVIDENCE OF FATIGUE DEVELOPS.

THE MAGNITUDE OF THE APPROXIMATE RATING LIFE IN HOURS, H, IS FOUND FROM THE FORMULA:

$$H = \frac{16667}{N}\left(\frac{C}{P}\right)^{\frac{10}{3}}$$

WHERE
C = BASIC DYNAMIC LOAD RATING
P = EQUIVALENT LOAD (COMBINED RADIAL AND SHAFT LOCATION LOAD)
N = REVOLUTIONS PER MINUTE

AVERAGE LIFE IS APPROXIMATELY FIVE TIMES THE RATING LIFE.

BASIC STATIC LOAD RATING: BASIC STATIC LOAD RATING IS THE MAXIMUM STATIC RADIAL LOAD WHICH WILL PRODUCE A PERMANENT DEFORMATION IN THE RACE AND ROLLERS, COMBINED AT ANY ONE CONTACT POINT, OF NOT MORE THAN .0001 OF THE ROLLER DIAMETER.

OPERATING TEMPERATURE: RECOMMENDED OPERATING TEMPERATURE NOT TO EXCEED 250 DEGREES FAHRENHEIT.

LIMITING RPM: LIMITING RPM IS THE RECOMMENDED SAFE DESIGN SPEED FOR NORMAL OPERATING CONDITIONS WITH GREASE LUBRICATION (INNER RING ROTATING).

NOTES:
(a) THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERRED HEREIN.
(b) REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BID.

APPROVED 30 APR 1959 REVISED

P.A. ORD
Other Cust
CE
USAF

PROCUREMENT SPECIFICATION
FF-B-185

TITLE
BEARINGS, ROLLER, CYLINDRICAL, SINGLE RO. OF ROLLERS, TWO ROLLER RETAINING RIBS ON INNER RING, TWO ROLLER RETAINING RIBS ON OUTER RING, ONE DIRECTION LOCATING...

SUPERSEDES:

MILITARY STANDARD

MS 19078

SHEET 3 OF 3

DD FORM 1 SEP 54 672-1